Triangles (Set A) — MCQ Worksheet

Mathematics • Chapter 6 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Two figures are congruent if they have:

(A) Same shape and size
(B) Same shape, may differ in size
(C) Same area only
(D) Same colour

Q2. Two figures are similar if they have:

(A) Same shape and size
(B) Same shape, may differ in size
(C) Same area
(D) Same perimeter only

Q3. All circles are:

(A) Congruent
(B) Similar
(C) Both
(D) Neither

Q4. All squares are:

(A) Congruent
(B) Similar
(C) Both
(D) Neither

Q5. In two similar triangles, corresponding angles are:

(A) Unequal
(B) Equal
(C) Supplementary
(D) Right angles only

Q6. The Basic Proportionality Theorem (BPT/Thales) states: a line drawn parallel to one side of a triangle divides the other two sides:

(A) In equal halves
(B) Proportionally
(C) At right angle
(D) Randomly

Q7. The criterion for similarity AAA states that if three angles of one triangle equal three angles of another, the triangles are:

(A) Congruent
(B) Similar
(C) Equilateral
(D) Right-angled

Q8. The SSS criterion for similarity is when corresponding sides are:

(A) Equal
(B) Proportional
(C) Equal in length only
(D) Perpendicular

Q9. The SAS criterion for similarity is when one angle is equal and:

(A) The sides containing that angle are proportional
(B) All sides are equal
(C) Hypotenuse equal
(D) Areas equal

Q10. Pythagoras theorem applies to:

(A) Equilateral triangles only
(B) Right-angled triangles
(C) Isosceles triangles
(D) Scalene triangles

Q11. In a right triangle with legs 3, 4 the hypotenuse is:

(A) 5
(B) 6
(C) 7
(D) $\sqrt{7}$

Q12. In a right triangle with legs 5, 12 the hypotenuse is:

(A) 13
(B) 17
(C) 15
(D) 12.5

Q13. If the sides of a triangle are 6, 8, 10, the triangle is:

(A) Equilateral
(B) Right-angled
(C) Obtuse
(D) Acute

Q14. Ratio of areas of two similar triangles equals:

(A) Ratio of corresponding sides
(B) Square of ratio of corresponding sides

(C) Cube of ratio of sides
(D) No relation

Q15. If ratio of corresponding sides of two similar triangles is 2:3, ratio of areas is:

(A) 2:3
(B) 4:9
(C) 8:27
(D) 6:9

Q16. Two triangles are similar with ratio of areas 16:25. Ratio of corresponding sides is:

(A) 16:25
(B) 4:5
(C) 2:5
(D) 8:25

Q17. In ΔABC, DE ■ BC; if AD = 2, DB = 3, AE = 4, then EC =

(A) 3
(B) 6
(C) 5
(D) 8

Q18. In a triangle ABC, if D and E are midpoints of AB and AC respectively, DE is:

(A) Equal to BC
(B) Half of BC
(C) Equal to AB
(D) Equal to AC

Q19. The triangle formed by midpoints of sides of an equilateral triangle is:

(A) Right-angled
(B) Equilateral
(C) Scalene
(D) Isosceles only

Q20. The triangles in two similar triangles with sides in ratio 1:2 have perimeters in ratio:

(A) 1:2
(B) 1:4
(C) 1:8
(D) 2:1

Q21. In ΔABC ~ ΔDEF, ∠A =40°, ∠B =60°. Then ∠F =

(A) 40°
(B) 60°
(C) 80°
(D) 120°

Q22. In a right triangle, if a perpendicular is drawn from the right angle to the hypotenuse, the two triangles formed are:

(A) Congruent to original
(B) Similar to original
(C) Equilateral
(D) Right-angled but not similar

Q23. The ratio of areas of two squares whose sides are in ratio 3:5 is:

(A) 3:5
(B) 9:25
(C) 6:10
(D) 27:125

Q24. Two triangles are said to be similar if they satisfy:

(A) AAA only
(B) SSS only
(C) Either AAA, SSS, or SAS criterion
(D) Only equal area

Q25. If the diagonals of a quadrilateral divide each other proportionally, the quadrilateral is a:

(A) Square
(B) Trapezium
(C) Kite
(D) Triangle

— End of Worksheet —